

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving
COURSE NAME: Cloud Computing and Big Data Analytics
COURSE CODE: CSA15

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2) Question Count: 5 Total Marks: 50

Q1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice. [10 Marks]

Answer:

A hypervisor is the virtualization layer that allows multiple virtual machines to share physical hardware. Type-1 (bare-metal) hypervisors such as VMware ESXi, Microsoft Hyper-V, and Xen run directly on the physical hardware, while Type-2 (hosted) hypervisors such as VMware Workstation, VirtualBox, and Parallels run on top of a conventional host operating system.

Comparison

Criteria	Type-1 Hypervisor (Bare-metal)	Type-2 Hypervisor
Performance	Near-native performance since it accesses hardware directly with minimal overhead	Lower performance
Security	Smaller attack surface; no underlying general-purpose OS to exploit; strong isolation between VMs	Larger attack surface
Manageability	Centralized enterprise management (e.g., vCenter), supports clustering, live migration, HA/DR	Managed like a normal OS
Scalability	Designed for datacenter-scale consolidation of hundreds of VMs	Suited for small-scale
Typical Use	Production servers, cloud datacenters	Developer sandboxes

Justification

For a cloud datacenter hosting mission-critical applications, the Type-1 hypervisor is the better choice. Mission-critical workloads demand maximum performance, minimal downtime, and strong isolation among tenants; since a Type-1 hypervisor eliminates the host-OS layer, it reduces latency, improves I/O throughput, and shrinks the exploitable attack surface. It also provides enterprise features like live migration, high availability, resource pooling, and centralized orchestration that are essential for large-scale, always-on cloud services. Type-2 hypervisors, being dependent on a host OS, introduce a single point of failure and performance bottleneck that is unacceptable for production-grade, mission-critical cloud environments.

Q2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces. [10 Marks]

Answer:

In a multi-tenant cloud environment, virtualization provides logical isolation between workloads that share the same physical hardware. The hypervisor creates separate, independent virtual machines for finance, healthcare, and student-portal workloads, each with its own virtual CPU, memory, storage, and network interfaces, so that one tenant cannot directly access another tenant's data or processes.

How Virtualization Supports Isolation

- **Compute isolation:** Each VM gets dedicated virtual CPU cycles and memory address space managed by the hypervisor scheduler, preventing one tenant's process from reading another tenant's memory.
- **Storage isolation:** Virtual disks (VMDK/VHD) are logically separated and mapped to different storage volumes, often with encryption keys unique to each tenant.
- **Network isolation:** Virtual LANs (VLANs), virtual switches, and network virtualization (e.g., VXLAN, software-defined networking) segment traffic so that finance, healthcare, and student-portal VMs cannot sniff or intercept each other's packets.
- **Policy-based access control:** Role-based access control (RBAC) and hypervisor-level security policies restrict administrative access based on tenant sensitivity, particularly important for regulated data such as healthcare (HIPAA) and finance (PCI-DSS).

Two Major Attack Surfaces

- **Hypervisor/VM-escape vulnerabilities:** If an attacker compromises the hypervisor itself or exploits a VM-escape flaw, they can break isolation and gain access to co-located VMs belonging to other tenants (e.g., moving from the student-portal VM into the healthcare VM).
- **Shared physical resource side channels:** Because tenants share physical CPU caches, memory buses, and storage I/O paths, side-channel attacks (such as cache-timing attacks) can leak sensitive data across VM boundaries even without direct hypervisor compromise.

Q3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks. [10 Marks]

Answer:

Snapshot Analysis

Metric	Observed Value	Healthy Range	Status
CPU Utilization	92%	< 75-80%	Critical - overloaded
Memory Utilization	88%	< 80%	High - near saturation
Storage I/O Latency	35 ms	< 10-15 ms	High - degraded storage
Average VM Boot Time	170 s	< 60 s	Poor - abnormally slow

Likely Bottlenecks

- **CPU contention:** 92% utilization indicates the physical host is CPU-bound; too many vCPUs are likely over-committed relative to physical cores, causing scheduling queues (CPU ready time) to grow, which directly slows down every VM including boot processes.
- **Memory pressure:** 88% memory usage is close to saturation, which can trigger hypervisor-level memory reclamation techniques such as ballooning or swapping to disk; swapping is extremely slow and compounds performance problems.
- **Storage I/O bottleneck:** A 35 ms latency is well above the acceptable 10-15 ms threshold, suggesting disk contention, insufficient IOPS provisioning, or an overloaded SAN/NAS backend; this directly explains the abnormally long 170-second boot time, since boot involves heavy disk read activity.
- **Compounding effect:** High CPU and memory pressure combined with slow storage creates a cascading bottleneck: VMs wait longer for CPU cycles, then stall further waiting on slow disk reads during OS initialization.

Recommended Corrective Actions

- Right-size VMs and reduce vCPU over-commitment ratio; migrate some VMs to another host to balance CPU load (use DRS/vMotion or live migration).
- Increase physical RAM or reduce memory over-allocation; enable memory compression rather than relying on disk-based swapping.
- Upgrade to faster storage (SSD/NVMe) or use storage tiering; implement storage I/O caching and increase provisioned IOPS.
- Enable resource monitoring and set automated alerts/thresholds (e.g., vSphere DRS, Hyper-V Resource Metering) to proactively rebalance workloads before saturation occurs.
- Consider horizontal scaling by adding another host to the cluster to distribute the workload across more physical resources.

Q4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations. [10 Marks]

Answer:

A Software Defined Datacenter (SDDC) abstracts all datacenter infrastructure -- compute, storage, and networking -- into software-controlled, policy-driven resources that can be provisioned, scaled, and managed programmatically, unlike a traditional datacenter where these resources are tightly coupled to fixed, manually configured hardware.

Compute Virtualization

In a traditional datacenter, applications run on dedicated physical servers, so provisioning a new application requires manual hardware procurement and installation, often taking days or weeks. In an SDDC, compute is virtualized through hypervisors, allowing new virtual machines or containers to be spun up in minutes via automated templates, enabling rapid scaling up or down based on demand.

Storage Virtualization

Traditional datacenters use storage that is physically bound to specific servers or arrays, making reallocation slow and rigid. SDDC pools all storage into a software-defined storage layer (e.g., VMware vSAN, software-defined storage arrays) that can be dynamically allocated, tiered, and replicated across the datacenter through policy-based automation, improving flexibility and reducing wasted capacity.

Network Virtualization

Traditional networks rely on physical switches, routers, and manual VLAN configuration, which is slow to reconfigure and error-prone. SDDC uses network virtualization technologies (e.g., VMware NSX, software-defined networking/SDN) to create virtual networks, firewalls, and load balancers entirely in software, allowing network topology and security policies to be reconfigured instantly without touching physical cabling.

Overall Agility Improvement

Because compute, storage, and network resources in an SDDC are abstracted, pooled, and centrally orchestrated through automation and APIs, organizations can provision entire application environments on demand, scale elastically, and recover from failures faster. This results in significantly greater agility, reduced operational cost, and faster time-to-market compared to the rigid, manually managed infrastructure of a traditional datacenter.

Q5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts. [10 Marks]

Answer:

Analysis of the Problem

In a multi-cloud federation, an organization uses services from multiple independent cloud providers (e.g., AWS, Azure, GCP) that must recognize and trust a common set of user identities. Identity mismatches occur when each provider maintains its own separate identity store, authentication protocol, or attribute format, so a user authenticated on one provider is not automatically recognized or trusted by another. This breaks single sign-on (SSO), causes inconsistent access control, creates duplicate/orphaned accounts, and introduces security gaps where privilege levels differ across providers for the same user.

Root Causes

- Lack of a common identity standard or protocol between providers (e.g., one uses SAML, another uses OAuth/OIDC without proper mapping).
- No centralized identity provider (IdP) acting as the trusted source of truth.
- Inconsistent attribute schemas (roles, groups, permissions) across provider domains.
- Absence of trust agreements (federation metadata, certificates) between the providers' identity systems.

Proposed Secure Identity and Privacy Design

- **Centralized Identity Federation using a trusted IdP:** Deploy a single, centralized Identity Provider (e.g., Azure AD, Okta, Keycloak) that authenticates users once and issues federated tokens trusted by all participating cloud providers, using standard protocols such as SAML 2.0 or OpenID Connect (OIDC)/OAuth 2.0.
- **Single Sign-On (SSO) with token-based trust:** Use signed security tokens (SAML assertions or JWTs) exchanged between the IdP and each cloud provider, so identity and role attributes are consistently interpreted across all domains, eliminating duplicate accounts.
- **Attribute mapping and standardization:** Define a common schema for user attributes (role, department, clearance level) and map each provider's local attributes to this standard during federation setup, ensuring consistent access-control decisions everywhere.
- **Mutual trust establishment:** Exchange federation metadata and digital certificates between the IdP and each cloud provider to cryptographically verify token authenticity and prevent identity spoofing.
- **Privacy-preserving design:** Apply the principle of minimal disclosure by sharing only the attributes each provider needs (attribute-based access control) rather than full user profiles, and encrypt identity tokens in transit (TLS) to protect personally identifiable information (PII).
- **Continuous governance:** Implement centralized identity lifecycle management (provisioning/de-provisioning via SCIM) and periodic access audits so identities remain consistent and secure as users join, change roles, or leave the organization.

This federated identity architecture ensures that a user is authenticated once and securely, consistently recognized across all cloud providers, resolving the identity mismatch problem while preserving privacy and strengthening overall security posture.

Total Marks Awarded: 50 / 50